# Answers and talk notes

Understand Ratios

### What your student is learning

Tonight we make a ratio bigger without changing what it compares. Multiply BOTH quantities by the same number — the scale factor — and you get an equivalent ratio.

## Answers

Ask your student to explain before you look.

1. **B.** 8:12. 2 × 4 = 8 and 3 × 4 = 12, so the new ratio is 8:12.
2. **C.** 4. 3 × 4 = 12 and 5 × 4 = 20, so the scale factor is 4.
3. **A.** 4:12. 1 × 4 = 4 and 3 × 4 = 12, so 4:12 is equivalent to 1:3.
4. **D.** 15. The flour went from 2 to 6, so the scale factor is 3. Milk: 5 × 3 = 15 cups.
5. **B.** 3. 6 ÷ 3 = 2 and 9 ÷ 3 = 3, so divide both by 3.
6. **B.** Yes, 5 × 3 = 15 and 8 × 3 = 24. Yes, 5 × 3 = 15 and 8 × 3 = 24, so both numbers use the scale factor 3.

### Talk about it

What number did you multiply by, and did you do the same thing to both parts?

### Home mission

Use dry beans as spoons. Put 1 bean for vinegar and 3 beans for oil. Ask your student to make the dressing with 4 beans of vinegar. How many oil beans? Then ask: what did you do to both piles?

### When they are stuck, ask

- What number was the first quantity multiplied by?
- Now do exactly that to the other quantity.
- Can you divide both new numbers back down to the ratio you started with?

### Try not to say

- Just add the same amount to both numbers.
- Close enough — a little extra of one will not matter.
- Only the first number has to change.